

let $L = L_0 \cup \{a_i\}_{i \in \mathbb{R}}$

$T = G \cup \{ \forall x (x =$

$$S \leq G$$

$$\frac{N \cap G}{(S \cap N)/N} \cong G/(S \cap N)$$

let G be a t. $G/[G, G]$ is abelian, G is countable and $G \leq H$ ~~is dense~~ where H is t.d.l.c. n.c.

$$\begin{array}{ccc} H & \longrightarrow & H/[G, G] \\ \uparrow & & \uparrow \\ G & \longrightarrow & G/[G, G] \end{array}$$

$$\frac{G \leq H}{[G, G] \leq H}$$

$$\Rightarrow \frac{G/[G, G]}{[G, G]} \cong \frac{G}{[G, G]}$$

$\Rightarrow H/[G, G]$ is abelian $\Rightarrow$ is elementary

is $[G, G]$ elementary?

yes, cause $[G, G]$ is abelian

$\hookrightarrow$ does solvable ^{dense subgroup} implies that the group is solvable

$\hookrightarrow$ we already have that solvable dense $\Rightarrow$ elementary

$\rightarrow$ locally finite & bounded order

$G \not\rightarrow G \Rightarrow f$ is iso

let G be countable / Hopfian and $H \supseteq G$ s.t. G is dense in H $\rightarrow$ t.d.l.c.n.c.

let $h = \lim_{n \in \mathbb{N}} g_n, g_n \in G$.

* $G \subseteq H, N \trianglelefteq G \Rightarrow N \trianglelefteq H$
 $\downarrow$
 G is dense even though $N \not\trianglelefteq H$

→ Commensurates no -

→ Proper Hopfian $\rightarrow$ embeddings into elementary grps

* $G \cong G$ then $\text{Ker}(\pi)$ has finite rank.
 $\downarrow$
comp. gen. element.

* If group with no proper commensurated subgroup $\rightarrow$ embed into discrete
 $\rightarrow$ finding what make the embedding elementary.